

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #6

Activity Name: ADC Module Interfacing

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Activity Date:	18-08-26
Submission Date:	18-08-26

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	10	

Faculty In-charge

Write an embedded C program for ARM architecture to drive ADC module. Debug and verify the program using IDE on a development computer system.

CODE:

```
#include <LPC213x.h>

#define RS (1 << 16)
#define EN (1 << 17)

unsigned int adc_value;

void delay(unsigned int count)
{
    unsigned int i, j;

    for(i = 0; i < count; i++)
        for(j = 0; j < 100; j++);
}

/* Send command to LCD */
void lcd_cmd(unsigned char cmd)
{
    IOCLR0 = RS;

    /* Send higher nibble */
    IOCLR0 = 0x003C0000;    // Clear P0.18 to P0.21
    IOSET0 = ((cmd & 0xF0) << 14);

    IOSET0 = EN;
    delay(10);
    IOCLR0 = EN;

    /* Send lower nibble */
    IOCLR0 = 0x003C0000;
    IOSET0 = ((cmd & 0x0F) << 18);

    IOSET0 = EN;
    delay(10);
    IOCLR0 = EN;
}

/* Send data to LCD */
void lcd_data(unsigned char data)
{
    IOSET0 = RS;

    /* Send higher nibble */
    IOCLR0 = 0x003C0000;
```

```

IOSET0 = ((data & 0xF0) << 14);

IOSET0 = EN;
delay(10);
IOCLR0 = EN;

/* Send lower nibble */
IOCLR0 = 0x003C0000;
IOSET0 = ((data & 0x0F) << 18);

IOSET0 = EN;
delay(10);
IOCLR0 = EN;
}

/* Display string */
void lcd_string(char *str)
{
    while(*str)
    {
        lcd_data(*str);
        str++;
    }
}

/* Display ADC number */
void lcd_number(unsigned int num)
{
    unsigned char thousands, hundreds, tens, ones;

    thousands = num / 1000;
    num = num % 1000;

    hundreds = num / 100;
    num = num % 100;

    tens = num / 10;
    ones = num % 10;

    if(thousands)
        lcd_data(thousands + '0');

    if(thousands || hundreds)
        lcd_data(hundreds + '0');

    if(thousands || hundreds || tens)

```

```

    lcd_data(tens + '0');

    lcd_data(ones + '0');
}

/* Initialize LCD */
void lcd_init(void)
{
    delay(100);

    lcd_cmd(0x02);    // 4-bit mode
    lcd_cmd(0x28);    // 2 lines, 5x7 matrix
    lcd_cmd(0x0C);    // Display ON, cursor OFF
    lcd_cmd(0x06);    // Increment cursor
    lcd_cmd(0x01);    // Clear display

    delay(100);
}

int main(void)
{
    /* LCD pins P0.16 to P0.21 as GPIO */
    PINSEL1 = 0x01000000;

    /* P0.16 to P0.21 as output */
    IODIR0 |= 0x003F0000;

    lcd_init();

    lcd_cmd(0x80);
    lcd_string("ADC VALUE:");

    while(1)
    {
        /* Configure and enable ADC0 Channel 1 */
        AD0CR = (1 << 1) | (10 << 8) | (1 << 21);

        /* Start ADC conversion */
        AD0CR |= (1 << 24);

        /* Wait for conversion */
        while(!(AD0GDR & (1UL << 31)));

        /* Extract 10-bit ADC value */
        adc_value = (AD0GDR >> 6) & 0x3FF;
    }
}

```

```
/* Move to second line */
```

```
lcd_cmd(0xC0);
```

```
/* Display ADC value */
```

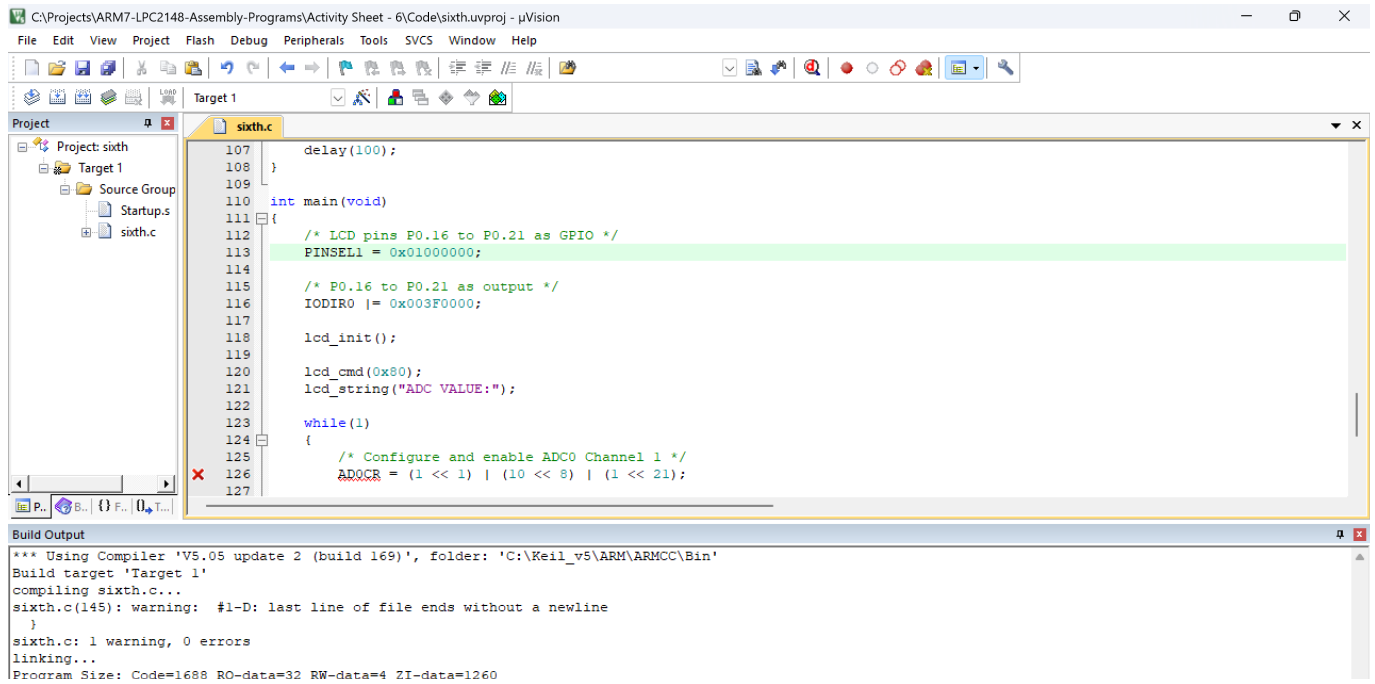
```
lcd_number(adc_value);
```

```
delay(500);
```

```
}
```

```
}
```

OUTPUT:



The screenshot displays the Keil uVision IDE interface. The main window shows the source code for `sixth.c`. The code includes a delay function call, a `main` function, and a `while` loop for ADC reading. The build output window at the bottom shows the compilation process, including the compiler version (V5.05 update 2), the target (Target 1), and the compilation of `sixth.c`. A warning is displayed: `sixth.c(145): warning: #1-D: last line of file ends without a newline`. The final program size is reported as Code=1688, RO-data=32, RW-data=4, and ZI-data=1260.

```
107     delay(100);
108 }
109
110 int main(void)
111 {
112     /* LCD pins P0.16 to P0.21 as GPIO */
113     PINSEL1 = 0x01000000;
114
115     /* P0.16 to P0.21 as output */
116     IODIR0 |= 0x003F0000;
117
118     lcd_init();
119
120     lcd_cmd(0x80);
121     lcd_string("ADC VALUE:");
122
123     while(1)
124     {
125         /* Configure and enable ADC0 Channel 1 */
126         ADC0CR = (1 << 1) | (10 << 8) | (1 << 21);
127     }
```

Build Output

```
*** Using Compiler 'V5.05 update 2 (build 169)', folder: 'C:\Keil_v5\ARM\ARMCC\Bin'
Build target 'Target 1'
compiling sixth.c...
sixth.c(145): warning: #1-D: last line of file ends without a newline
}
sixth.c: 1 warning, 0 errors
linking...
Program Size: Code=1688 RO-data=32 RW-data=4 ZI-data=1260
```

